

corrigé zakito

Énoncé

Autocorrection A

- (i) $p \in \mathbb{R}^2 \Rightarrow B(p, 1) \subset \mathbb{R}^2$.
- (ii) $p \neq 0, \quad r = \|p\|/2 > 0, \quad \|z - p\| < r \Rightarrow \|z\| > \|p\|/2$.
- (iii) $p = (a, b) \in (0, 1)^2, \quad r = \frac{1}{2} \min(a, 1 - a, b, 1 - b) > 0$
 $\Rightarrow B(p, r) \subset (0, 1)^2$.
- (iv) $U = h^{-1}((0, +\infty)), \quad h(x, y) = y - \cos x \in C^0(\mathbb{R}^2)$.

Autocorrection B

$$q = p + (r, 0), \quad q \in D(p, r), \quad q + (\varepsilon/2, 0) \in B(q, \varepsilon) \setminus D(p, r). \\ \forall \varepsilon > 0, \quad B(q, \varepsilon) \not\subset D(p, r).$$

Exercice 1

1

$$p \in \bigcup_i U_i \Rightarrow \exists i, p \in U_i \Rightarrow \exists r > 0, B(p, r) \subset U_i \subset \bigcup_i U_i.$$

2

$$p \in U \cap V, \quad B(p, r) \subset U, \quad B(p, s) \subset V \Rightarrow B(p, \min(r, s)) \subset U \cap V.$$

3

$$U_n = \pi_2^{-1}((-1/n, 1/n)), \quad \pi_2(x, y) = y \in C^0. \\ \bigcap_{n \geq 1} U_n = \mathbb{R} \times \{0\}, \quad (a, \varepsilon/2) \in B((a, 0), \varepsilon) \setminus \bigcap_n U_n.$$

Exercice 2

$$B(p, r) \subset U, \quad g(t) = f(p + \frac{r}{2}(\cos t, \sin t)) - f(p - \frac{r}{2}(\cos t, \sin t)). \\ g \in C^0([0, \pi]), \quad g(\pi) = -g(0) \Rightarrow \exists t_0 \in [0, \pi], \quad g(t_0) = 0. \\ p + \frac{r}{2}(\cos t_0, \sin t_0) \neq p - \frac{r}{2}(\cos t_0, \sin t_0) \Rightarrow f \text{ non injective.}$$

Exercice 3

1(a)

$$K = S \times T, \quad p_n = (x_n, y_n) \in K \Rightarrow \exists (n_k), \quad x_{n_k} \rightarrow x \in S, \quad y_{n_k} \rightarrow y \in T. \\ |f(p_n)| \geq n \Rightarrow |f(p_{n_k})| \rightarrow +\infty, \quad f(p_{n_k}) \rightarrow f(x, y), \quad \perp. \\ M = \sup_K f \in \mathbb{R}, \quad M - 1/n < f(p_n) \leq M. \\ p_{n_k} \rightarrow p \in K \Rightarrow f(p) = \lim_k f(p_{n_k}) = M. \\ m = \inf_K f \in \mathbb{R}, \quad m \leq f(q_n) < m + 1/n, \quad q_{n_k} \rightarrow q \in K \Rightarrow f(q) = m.$$

1(b)

$$\neg \text{continuité uniforme} \Rightarrow \exists \varepsilon > 0, (p_n), (q_n) \in K^{\mathbb{N}}, \quad \|p_n - q_n\| \rightarrow 0, \quad |f(p_n) - f(q_n)| \geq \varepsilon. \\ p_{n_k} \rightarrow p \in K \Rightarrow q_{n_k} \rightarrow p \Rightarrow |f(p_{n_k}) - f(q_{n_k})| \rightarrow 0, \quad \perp.$$

2

$$|F(x) - F(a)| \leq \sup_{t \in [0,1]} |f(x,t) - f(a,t)| \xrightarrow{x \rightarrow a} 0$$

par continuité uniforme sur $[a-1, a+1] \times [0, 1]$.

Autocorrection C

	∂_x	∂_y
f_1	$1/y$	$-x/y^2$
f_2	$-1/(x^2 y^2)$	$-2/(xy^3)$
f_3	$\ln(xy) + 1$	x/y
f_4	$e^{-x}(2x \cos(x^2 + y^2) - \sin(x^2 + y^2))$	$2ye^{-x} \cos(x^2 + y^2)$

Autocorrection D

$$\frac{N(h, 0) - N(0, 0)}{h} = \frac{|h|}{h}, \quad \frac{N(0, h) - N(0, 0)}{h} = \frac{|h|}{h}.$$

$$\partial_x N(x, y) = \frac{x}{\sqrt{x^2 + y^2}}, \quad \partial_y N(x, y) = \frac{y}{\sqrt{x^2 + y^2}} \quad ((x, y) \neq 0).$$

Ces deux fonctions sont continues sur $\mathbb{R}^2 \setminus \{0\}$.

Autocorrection E

$$\theta \in C^1(\mathbb{R}), \quad \theta'(t) = 2t \partial_1 f(t^2, t^3) + 3t^2 \partial_2 f(t^2, t^3).$$

Autocorrection F

$$\partial_x u_1(x, y) = \partial_2 f(y, x), \quad \partial_y u_1(x, y) = \partial_1 f(y, x),$$

$$u'_2(x) = \partial_1 f(x, x) + \partial_2 f(x, x),$$

$$\partial_x u_3(x, y) = \partial_2 f(y, f(x, x))[\partial_1 f(x, x) + \partial_2 f(x, x)],$$

$$\partial_y u_3(x, y) = \partial_1 f(y, f(x, x)),$$

$$u'_4(x) = \partial_1 f(x, f(x, x)) + \partial_2 f(x, f(x, x))[\partial_1 f(x, x) + \partial_2 f(x, x)].$$

Les compositions et les dérivées affichées sont continues.

Exercice 4

$$H(t) = \int_0^t f(s) ds \in C^1, \quad \varphi(x, y) = H(y) - H(x).$$

$$\partial_x \varphi = -f(x), \quad \partial_y \varphi = f(y), \quad \varphi \in C^1(\mathbb{R}^2).$$

Exercice 5

1

$$p \neq 0 \Rightarrow f \text{ de classe } C^1 \text{ au voisinage de } p \Rightarrow D_v f(p) = \nabla f(p) \cdot v.$$

$$v = (a, b), \quad b \neq 0 : \quad \frac{f(ta, tb)}{t} = \frac{a^2 b}{t^2 a^4 + b^2} \rightarrow \frac{a^2}{b}.$$

$$b = 0 \Rightarrow f(ta, 0) = 0 \Rightarrow D_{(a,0)} f(0) = 0.$$

2

$$t \neq 0 \Rightarrow f(t, t^2) = \frac{1}{2} \not\rightarrow f(0, 0) = 0.$$

Exercice 6

$$\begin{aligned}\partial_x(f + ig) &= P'(x + iy), & \partial_y(f + ig) &= iP'(x + iy). \\ P'(x + iy) &= u + iv \Rightarrow f_x = u = g_y, & f_y &= -v = -g_x.\end{aligned}$$

Exercice 7**1**

$$\begin{aligned}h(t) &= f(p + t(q - p)), & h'(t) &= \nabla f(p + t(q - p)) \cdot (q - p) = 0. \\ f(q) &= h(1) = h(0) = f(p).\end{aligned}$$

2

$$\begin{aligned}U &= \{(x, y) : x \neq 0\}, & f(x, y) &= \begin{cases} 0 & x < 0, \\ 1 & x > 0. \end{cases} \\ f &\in C^1(U), & \nabla f &= 0, & f(-1, 0) &\neq f(1, 0).\end{aligned}$$

Exercice 8

$$\begin{aligned}f_y = 0 &\Rightarrow \forall x, y, & f(x, y) &= f(x, 0) = h(x), & h &\in C^1, & h'(x) &= f_x(x, 0). \\ f(x, y) &= h(x) &\Rightarrow f_y(x, y) &= 0.\end{aligned}$$

Exercice 9

Définition corrigée : $g(r, \theta) = f(r \cos \theta, r \sin \theta)$.

1

$$\begin{aligned}g_r &= \cos \theta f_x(r \cos \theta, r \sin \theta) + \sin \theta f_y(r \cos \theta, r \sin \theta), \\ g_\theta &= -r \sin \theta f_x(r \cos \theta, r \sin \theta) + r \cos \theta f_y(r \cos \theta, r \sin \theta).\end{aligned}$$

2

$$f \text{ radiale} \iff \forall r \geq 0, g(r, \cdot) \text{ constante} \iff g_\theta = 0 \iff x f_y - y f_x = 0.$$

Exercice 10

$$f(tp) = t^k f(p) \Rightarrow \left. \frac{d}{dt} f(tp) \right|_{t=1} = p \cdot \nabla f(p) = k f(p).$$

Réciproquement, pour $p \neq 0$ et $t > 0$,

$$\begin{aligned}\frac{d}{dt} (t^{-k} f(tp)) &= t^{-k-1} ((tp) \cdot \nabla f(tp) - k f(tp)) = 0. \\ t^{-k} f(tp) &= f(p).\end{aligned}$$

Exercice 11**1**

$$\begin{aligned}F(u, v) &= f(u, v - 2u), & F_u &= f_x(u, v - 2u) - 2f_y(u, v - 2u). \\ f_x - 2f_y &= 0 \iff F(u, v) = h(v) \iff f(x, y) = h(2x + y), & h &\in C^1(\mathbb{R}).\end{aligned}$$

2

$$\begin{aligned}(\partial_x - 2\partial_y) \frac{x^2}{2} &= x, \\ f(x, y) &= \frac{x^2}{2} + h(2x + y), & h &\in C^1(\mathbb{R}).\end{aligned}$$

Exercice 12

1

$$\frac{f(a+h, t) - f(a, t)}{h} = \int_0^1 f_x(a+sh, t) ds.$$

$$\left| \frac{F(a+h) - F(a)}{h} - \int_0^1 f_x(a, t) dt \right| \leq \sup_{0 \leq s, t \leq 1} |f_x(a+sh, t) - f_x(a, t)| \rightarrow 0.$$

$$F'(a) = \int_0^1 f_x(a, t) dt, \quad F' \in C^0$$

par l'exercice 3 appliqué à f_x .

2(a)

$$\gamma'(x) = e^{-x^2}, \quad F'(x) = -2xe^{-x^2} \int_0^1 e^{-x^2 t^2} dt = -2e^{-x^2} \gamma(x).$$

La formule vaut aussi pour $x = 0$.

2(b)

$$(F + \gamma^2)' = 0, \quad F(0) = \int_0^1 \frac{dt}{1+t^2} = \frac{\pi}{4}, \quad \gamma(0) = 0.$$

$$F(x) = \frac{\pi}{4} - \gamma(x)^2.$$

2(c)

$$0 \leq F(x) \leq e^{-x^2} \int_0^1 \frac{dt}{1+t^2} = \frac{\pi}{4} e^{-x^2} \rightarrow 0.$$

$$x \geq 0 \Rightarrow \gamma(x) \geq 0, \quad \int_0^{+\infty} e^{-t^2} dt = \frac{\sqrt{\pi}}{2}.$$

Autocorrection G

1

$$\nabla f_1 = (-\sin x, 2y) = 0 \iff (x, y) = (k\pi, 0).$$

$$f_1 \geq -1, \quad f_1((2k+1)\pi, 0) = -1.$$

Les $((2k+1)\pi, 0)$ sont des minima globaux stricts localement.

$$f_1(2k\pi + t, 0) < 1 < f_1(2k\pi, t) \quad (0 < |t| < \pi).$$

Aucun maximum local ou global.

2

$$\partial_y f_2 = 2e^{3x}y + e^x = 0 \Rightarrow y = -\frac{1}{2}e^{-2x},$$

$$\partial_x f_2 = 3e^{3x}y^2 + e^xy = \frac{1}{4}e^{-x} > 0.$$

Aucun point critique, donc aucun extremum local ou global.

3

$$X = x - 1, \quad Y = y + 1, \quad f_3(x, y) = -8 + 3X^2 - 2XY + 3Y^2 = -8 + 2(X^2 + Y^2) + (X - Y)^2.$$

Minimum global unique : $f_3(1, -1) = -8$. Aucun maximum.

4

$$\nabla f_4 = f_4 \left(\arctan y, \frac{x}{1+y^2} \right) = 0 \iff (x, y) = (0, 0).$$

$$t \neq 0 : f_4(t, t) > 1 > f_4(t, -t).$$

Aucun extremum local ou global.

Exercice 13

1

$$f \in C^\infty, \quad \nabla f = (e^y + ye^x, xe^y + e^x).$$

$$\nabla f = 0 \Rightarrow x < 0, y < 0, xy = 1.$$

$$a = -x > 0, y = -1/a, \quad 1/a - a = \ln a.$$

$$H(a) = 1/a - a - \ln a, \quad H'(a) = -1/a^2 - 1 - 1/a < 0, \quad H(1) = 0.$$

$$\nabla f = 0 \iff (x, y) = (-1, -1).$$

2

$$\text{Hess } f(-1, -1) = e^{-1} \begin{pmatrix} -1 & 2 \\ 2 & -1 \end{pmatrix}.$$

$$f(-1+t, -1+t) = -2/e + t^2/e + o(t^2),$$

$$f(-1+t, -1-t) = -2/e - 3t^2/e + o(t^2).$$

Aucun extremum local.

Exercice 14

$$\nabla f(p) = 0, \quad v = q - p, \quad h(t) = f(p + tv).$$

$$t > 0 : \quad th'(t) = \langle \nabla f(p + tv) - \nabla f(p), tv \rangle \geq 0.$$

$$f(q) - f(p) = \int_0^1 h'(t) dt \geq 0.$$

Exercice 15

1

$$f = y^2 - 3x^2y + 2x^4, \quad \nabla f = (-6xy + 8x^3, 2y - 3x^2).$$

$$\nabla f = 0 \Rightarrow y = \frac{3}{2}x^2, \quad -x^3 = 0 \Rightarrow (x, y) = (0, 0).$$

$$f(0, t) = t^2 > 0 = f(0, 0) \quad (t \neq 0).$$

2

$$v = (a, b), \quad f(tv) = t^2(b - ta^2)(b - 2ta^2).$$

$$b \neq 0 \Rightarrow (b - ta^2)(b - 2ta^2) \rightarrow b^2 > 0; \quad b = 0 \Rightarrow f(tv) = 2a^4t^4 \geq 0.$$

3

$$f(t, \frac{3}{2}t^2) = -\frac{1}{4}t^4 < 0 \quad (t \neq 0).$$

Aucun extremum local en $(0, 0)$.

Exercice 16

Énoncé impossible tel qu'imprimé.

$$\varphi : \mathbb{R}^2 \rightarrow (-1, 1), \quad \psi = \varphi^{-1}, \quad \psi \circ \varphi = \text{id}_{\mathbb{R}^2}$$

$$\Rightarrow D\psi(\varphi(p))D\varphi(p) = I_2 \Rightarrow 2 = \text{rg } I_2 \leq 1, \quad \perp.$$

Exercice 17

1

$$f(f(x, y), z) = f(x, f(y, z)).$$

Dérivation en $z = e$:

$$\partial_2 f(x * y, e) = \partial_2 f(x, y) \partial_2 f(y, e).$$

2

$$a(y) = \partial_2 f(y, e), \quad a(e) = 1.$$

$$x = y^{-1} \Rightarrow 1 = \partial_2 f(y^{-1}, y) a(y) \Rightarrow a(y) \neq 0.$$

$$a \in C^0(\mathbb{R}), \quad a(e) > 0 \Rightarrow \forall y, a(y) > 0.$$

3

$$\varphi(e) = 0, \quad \varphi(f(x, y)) = \varphi(x) + \varphi(y).$$

Dérivation en $y = e$:

$$\varphi'(x) a(x) = \varphi'(e) = c \Rightarrow \varphi(x) = c \int_e^x \frac{dt}{a(t)}.$$

4

$$c \neq 0, \quad \varphi'(x) = c/a(x) \neq 0.$$

$$\frac{d}{dy}(\varphi(f(x, y)) - \varphi(y)) = \frac{c \partial_2 f(x, y)}{a(x * y)} - \frac{c}{a(y)} = 0.$$

$$y = e \Rightarrow \varphi(x * y) = \varphi(x) + \varphi(y).$$

$$J = \varphi(\mathbb{R}) \text{ intervalle ouvert contenant } 0, \quad J + J \subset J, \quad -J = J.$$

$$(-\delta, \delta) \subset J \Rightarrow \forall n \geq 1, (-n\delta, n\delta) \subset J \Rightarrow J = \mathbb{R}.$$

$$\varphi \text{ strictement monotone, surjective,} \quad (\varphi^{-1})'(u) = \frac{a(\varphi^{-1}(u))}{c} \in C^0.$$

$$\varphi : (\mathbb{R}, *) \simeq (\mathbb{R}, +), \quad x * y = y * x.$$

Le sens de l'isomorphisme imprimé dans l'énoncé est inversé.

5

$$(u, v) * (s, t) = (u + s, v + e^u t).$$

$$((u, v) * (s, t)) * (a, b) = (u + s + a, v + e^u t + e^{u+s} b) = (u, v) * ((s, t) * (a, b)).$$

$$e = (0, 0), \quad (u, v)^{-1} = (-u, -e^{-u} v).$$

$$(1, 0) * (0, 1) = (1, e) \neq (1, 1) = (0, 1) * (1, 0).$$